

HINTS AND SOLUTIONS

Q.1 C

Sol. Same $q\Delta V$ so same KE but $m_e < m_p$ So $V_e > V_p$ this in velocity selection potential different must be increased.

Q.2 B

Q.3 B

Q.4 B

Q.5 D

Q.6 B

Q.7 D

Q.8 B

Q.9 C

Q.10 D

Sol. current inclosed is same so $\oint \mathbf{B} \cdot d\mathbf{\ell}$ will be same for both cases.

Q.11 B

Sol.
$$\frac{m}{3} \frac{v^2}{r} = \frac{k \left(\frac{2l^2}{9} \right)}{r^2} - \frac{k \frac{e^2}{9}}{(2r)^2} = \frac{ke^2}{9r^2} \left(2 - \frac{1}{4} \right) = \frac{7ke^2}{36r^2}$$
$$\Rightarrow v^2 = \frac{7ke^2}{12mr}$$

$$M = \frac{q\omega r^2}{2} \times 2$$

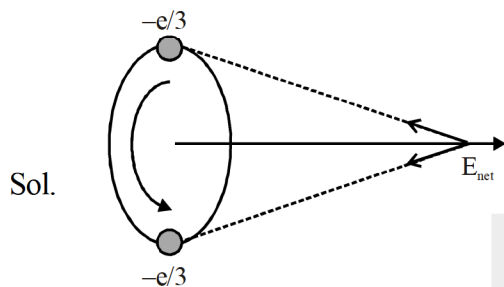
$$= \frac{e}{3} \times \frac{v}{r} \times r^2 = \frac{eh}{2\pi m}$$

$$r = \frac{3h}{2\pi m} \times v$$

$$\Rightarrow u^2 = \frac{7ke^2}{12m} \times 2\pi mv$$

$$u = \frac{7e^2}{48\pi \epsilon_0} \times \frac{2\pi}{3h} = \frac{7e^2}{72h \epsilon_0}]$$

Q.12 A



Q.13 A

Sol.
$$B = \frac{\mu_0 i r^2}{2(r^2 + x^2)^{3/2}} = \frac{\mu_0 \frac{2e}{3} \times \frac{\omega}{2\pi} r^2}{2(r^2 + x^2)^{3/2}}$$

$$E = \frac{k \times 2e/3}{x^2} - \frac{k \times 2e/3x}{(r^2 + x^2)^{3/2}} = \frac{2ek}{x^2} \left(1 - \frac{1}{\left(1 + \frac{r^2}{x^2} \right)^{3/2}} \right)$$

$$E = \frac{2ek}{3x^2} \left(1 - \left(1 - \frac{3}{2} \frac{r^2}{x^2} \right) \right) = \frac{ekr^2}{x^4} = \frac{er^2}{4\pi \epsilon_0 x^4}$$

$$B = \frac{\mu_e e \omega r^2}{6\pi x^3} = \frac{\mu_0 e r^2}{6\pi x^3} \times \frac{7e^2}{72h \epsilon_0 r}$$

$$\frac{E}{B} = \frac{108hr}{7\mu_0 e^2 x}$$

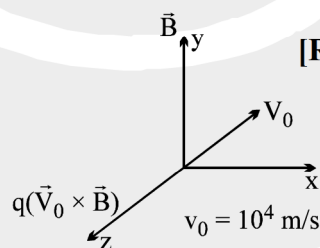
Q.14 AD

Q.15 B

Sol. $q\vec{v}_0 \times \vec{B} = q\vec{E}$

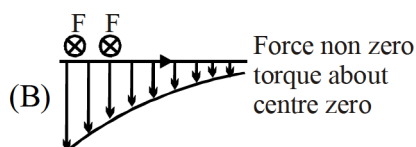
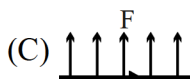
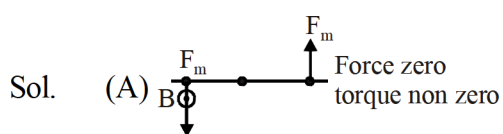
$$E = VB \sin 90^\circ = 10^4 \times 10^{-2} = 100 \text{ V/m}$$

$\vec{E}$ along $(-z)$ direction



[RT-7, XII, 02-12-07]

Q.16 [Ans. (A) P,S (B) R,S (C) Q,R]



Q.17 [Ans. $\frac{\mu_0}{8\pi} \sqrt{2} \frac{I}{a}$]

Q.18 [Ans. $i\ell^2 B \left(1 - \frac{4}{\pi}\right)$]

Q.19 Ans. $\omega = 2 \sqrt{\frac{\pi I B_0}{m}}$

Sol. $\vec{\mu}_1 = \pi R^2 I (-\hat{k})$

$U_1 = -\vec{\mu}_1 \cdot \vec{B} = 0$

$\vec{\mu}_2 = \pi R^2 I (\hat{i})$

$U_2 = -\vec{\mu}_2 \cdot \vec{B} = -\pi R^2 I B_0$

$\Delta U = \pi R^2 I B_0 = \frac{1}{2} \left[\frac{m R^2}{2} \right] \omega^2$

$\omega = 2 \sqrt{\frac{\pi I B_0}{m}}$

Q.20 Ans. $\vec{B} = \frac{\vec{B}}{|\vec{B}|} = \frac{4}{5} \hat{j} + \frac{3}{5} \hat{k}$

Sol. Let $\vec{B} = B_1 \hat{i} + B_2 \hat{j} + B_3 \hat{k}$

Hence $q[\vec{v} \times (B_1 \hat{i} + B_2 \hat{j} + B_3 \hat{k})] = qvB(-3\hat{j} + 4\hat{k})$

$\Rightarrow B_2 \hat{k} - B_3 \hat{j} = -3B_1 \hat{j} + 4B_1 \hat{k} \Rightarrow B_2 = 4B_1, B_3 = 3B_1$

Also $q[\vec{v} \times (B_1 \hat{i} + B_2 \hat{j} + B_3 \hat{k})] = qvB(3\hat{i})$

$\Rightarrow -B_2 \hat{k} + B_3 \hat{j} = 3B_1 \hat{i}$

$\Rightarrow B_1 = 0$

Hence $\vec{B} = 4B_1 \hat{j} + 3B_1 \hat{k}$

$\Rightarrow \vec{B} = \frac{\vec{B}}{|\vec{B}|} = \frac{4}{5} \hat{j} + \frac{3}{5} \hat{k}$

Q.21 [Ans. $\frac{\mu_0 i^2}{4\pi \sin \frac{\theta}{2} \tan \frac{\theta}{2} \ell g}$]

Sol. Separation $= 2\ell \sin \theta/2 = x$ (say)

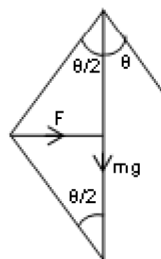
$\therefore \text{Force} = \frac{\mu_0 i^2 \ell}{2\pi x} = \frac{\mu_0 i^2 \ell}{2\pi (2\ell \sin \theta/2)}$

downward force $= mg = (\ell \lambda) g$

$\therefore \tan \theta/2 = \frac{F_E}{F_G} = \frac{\mu_0 i^2 \ell}{(4\mu \ell \sin \theta/2) \ell \lambda g}$

...(i)

...(ii)



Q.22 Ans. $I_0 = \frac{2\pi\rho_0}{3R} a^2(d-a)$

Sol. For conductor with current into the plane
for a circle of radius a in the cross-section of conductor current flowing within it

$$I = \int_0^a 2\pi x dx \cdot r_0(x/R)$$

$$= \frac{2\pi\rho_0}{R} \frac{a^3}{3}$$

For all point on the circle of radius a due to symmetry B is same. Applying Ampere's law

$$\int_0^{2\pi a} B \cdot d\ell = \mu_0 I \Rightarrow B_0 = \frac{\mu_0}{2\pi a} \frac{2\pi\rho_0}{R} \frac{a^3}{3}$$

At P , field B is $\perp$ to OP upward as shown

Now field B_0 due to wire must balance it, B_0 must be downward opposite to be B of same magnitude
 $\Rightarrow$ current in wire should also be into the plane of paper, Let it be I_0 .

$$B = \frac{\mu_0}{4\pi} \frac{2I_0}{d-a}$$

$$\vec{B} + \vec{B}_0 = 0$$

$$|\vec{B}_0| = |\vec{B}|$$

$$\Rightarrow \frac{\mu_0}{4\pi} \frac{2I_0}{d-a} = \frac{\mu_0}{2\pi} \frac{2\pi\rho_0}{R} \frac{a^2}{3}$$

$$\Rightarrow I_0 = \frac{2\pi\rho_0}{3R} a^2(d-a)$$

Q.23 [Ans. $F_1 = B^2 / 2\mu_0$]